

Small Language Models for Battle Decision-Making in Pokémon Blue

Structured prompting fixes action formatting, but not decision quality

Deliverable 1 • Generative Artificial Intelligence (580694), Spring 2026 • Universidad de Concepción

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Repository: **github.com/usuario/repositorio**

1. TASK SPECIFICATION

The model acts as the **battle-decision module** of a Pokémon Blue agent.

Input. A structured battle state: both Pokémon's type(s), HP% and Speed, plus the four available moves with *name*, *type*, *power*, *accuracy* and *remaining PP*.

Output. Exactly one executable label: MOVE_1, MOVE_2, MOVE_3 or MOVE_4.

What counts as correct. The label that maximises an explicit, deterministic offensive policy,

$$\text{score} = \text{power} \times \text{accuracy} \times \text{STAB} \times \text{type-effectiveness}$$

with $\text{score} = 0$ when $\text{PP} = 0$. The ground truth is therefore reproducible and auditable case by case.

Why it is non-trivial. The five attributes trade off: the highest-*power* move is optimal in only 19 of 30 states, and Gen I immunities (Electric → Ground, Normal → Ghost) zero out the strongest-looking option. The model must *combine* constraints, not retrieve a fact.

2. PILOT EXPERIMENT (V1)

Model. Qwen2.5-1.5B-Instruct, greedy decoding, $\text{max_new_tokens}=48$, seed 20260830, identical system prompt in both conditions.

Design. 30 synthetic battle states (10 EASY / 10 MEDIUM / 10 HARD, graded by the best-vs-second score ratio: $\geq 1.80 / 1.25\text{--}1.80 / 1.01\text{--}1.25$).

The same 30 states go to both prompts → **60 inferences**, offline, with no emulator inside the measurement loop.

Conditions. DIRECT = natural request ("What move would you use next?"). STRUCTURED = same state, explicit reasoning checklist, and a hard output contract (MOVE_N only).

Metrics. STRICT (exactly MOVE_N, executable) · INTERP (intent recoverable) · ACC (matches ground truth) · *regret* (share of the best score lost).

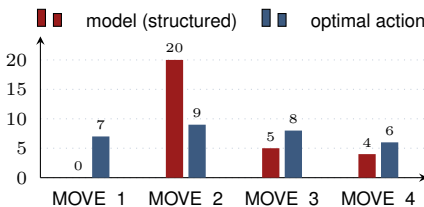
Prompt	STRICT	INTERP	ACC	EXEC-C.
DIRECT	0.0%	36.7%	3.3%	0.0%
STRUCTURED	96.7%	96.7%	16.7%	16.7%

Mean regret: DIRECT 0.41, STRUCTURED 0.49. $n = 30$ states per condition.

3. FAILURE DIAGNOSIS

Structured prompting almost eliminated the *interface* problem (STRICT 0% → 96.7%; one answer was still unexecutable), so the agent can now be driven automatically. It did *not* solve the task — decision quality stays **below chance**.

The model answers by position, not by state. Across the 29 parseable structured answers it returns MOVE_2 20 times and MOVE_1 *never*, while the optimal action is nearly uniform over the four slots ($\chi^2(3) = 31.8$, $p < 10^{-6}$ for its choice distribution against uniform).



It only scores when the answer happens to be MOVE_2. Hit rate by the position of the optimal move: MOVE_1 0/7, MOVE_2 4/9, MOVE_3 1/8, MOVE_4 0/6.

Trivial policies beat the model.

Policy	Accuracy
Qwen2.5-1.5B, structured prompt	16.7%
Uniform random over 4 moves	25.0%
Constant "always MOVE_2"	30.0%
$\max(\text{power} \times \text{accuracy})$, $\text{PP} > 0$	53.3%
$\max(\text{power})$ — ignores types entirely	63.3%

7. EXECUTION FEASIBILITY & REPRODUCIBILITY

The pilot **already ran end to end** on the team's own hardware: 60 inferences, **105.5 min** wall-clock on CPU, deterministic (greedy, fixed seed). Memory estimated as $\text{parameters} \times 2$ bytes (fp16) plus KV cache: all three fit in a free Colab T4 (16 GB), and Qwen2.5-1.5B is verified directly by the pilot. The repository ships the benchmark script, its four result artefacts and the PyBoy demo — kept *outside* the measurement loop, so emulator-control bugs can never be mistaken for model failures.

Repo: **github.com/usuario/repositorio**

Diagnosis

The main observed bottleneck is **not** output formatting but **multi-constraint action selection**: asked to weigh type effectiveness, STAB, power, accuracy and PP over four candidates at once, the 1.5B model collapses onto a **position prior** that is almost independent of the battle state. A one-line heuristic that ignores type charts altogether is **3.8×** more accurate.

4. IS THE GROUND TRUTH THE PROBLEM?

A low score against a hand-made policy could just mean the policy is wrong. Three checks say otherwise — these errors are indefensible under *any* rule:

- **Zero-effect move.** Case 16 (below): *Thunderbolt* (Electric) into a **Ground** opponent — immune, 0 damage.
- **Unexecutable move.** Case 29: a move with **PP = 0**; the agent physically cannot issue it.
- **Strictly worst option** chosen in **6 of 29** answers (21%) — the lowest-scoring of the four.

CASE 16 - player Flying (41% HP) vs Ground (53%)

MOVE_1	Psybeam	Psychic	P65	A100	score	65.0
MOVE_2	Thunderbolt	Electric	P95	A100	score	0.0
MOVE_3	Rock Throw	Rock	P50	A65	score	16.3
MOVE_4	Ember	Fire	P40	A100	score	40.0

ground truth = MOVE_1 model = **MOVE_2**

The model takes the highest *power* number and ignores the immunity that makes it useless.

No difficulty gradient. Structured accuracy is EASY 2/10, MEDIUM 1/10, HARD 2/10. Graded reasoning should degrade as cases get harder; a position prior should not — and it does not.

5. THREATS TO VALIDITY, AND THE V2 FIX

- **T1.** DIRECT never requested MOVE_N, so its 0% STRICT is true by construction; it is reported as an *interface baseline*, not as reasoning evidence.
- **T2.** $\text{max_new_tokens}=48$ truncated free-text DIRECT answers, deflating its 36.7% INTERP.
- **T3.** The reference policy is offensive-only: no status moves, switching, items or multi-turn play.

v2. Both prompts enforce the same MOVE_N contract, so the only difference is the reasoning scaffold; $n = 50$ states; every state replayed with its moves **permuted**, measuring the position prior directly.

6. PROPOSED MODELS (ALL FAR BELOW 8B)

Model	Par.	MMLU	IFEval	GSM8K	fp16
Qwen2.5-1.5B-Instr.	1.5B	50.7	42.5	73.2	3.1 GB
Llama-3.2-3B-Instr.	3.2B	63.4	77.4	77.7	6.4 GB
Phi-3.5-mini-instr.	3.8B	69.0	n.r.	86.2	7.6 GB

Sources: official model cards for Qwen2.5, Llama 3.2 and Phi-3.5-mini (full references in the repository). Protocols differ, so the figures place the models in a class rather than rank them. n.r. = not reported.

- **IFEval: instruction following.** A general benchmark for obeying constrained output formats, which is what the controller needs. Note that under the structured prompt Qwen already reaches 96.7% compliance despite its 42.5, so format is *not* what separates the candidates.
- **GSM8K: multi-step reasoning.** The closest public proxy to the step that does fail here, combining several numeric attributes. Phi-3.5-mini leads at **86.2** with only 3.8B: the direct test of whether that bottleneck lifts.

Qwen2.5-1.5B is also the baseline: the only candidate with a *measured* profile on this task. The task does not obviously require an 8B model, which motivates evaluating substantially smaller candidates; the largest here is **3.8B**.